

Khachatur Dallakyan

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EDUCATION

Karolinska Institute, Stockholm University	Sweden
Joint Master's Programme in Health Informatics	2024-2026
Awards: The Swedish Institute Scholarship for Global Professionals	
The Chinese University of Hong Kong	Hong Kong
Bachelor of Sc. in Artificial Intelligence: Systems and Technologies (GPA: 3.29 / 4.00)	2020-2024
Awards: Second class upper division with Honours, Belt and Road Scholarship	
International Baccalaureate Diploma Programme at Quantum College	Armenia
Total points: 37/45	2018-2020
Subjects: Mathematics, Computer Science, ITGS, Armenian, Economics, English.	

RESEARCH AND WORK EXPERIENCE

Exploring the Enteric Nervous System (ENS) through Single-Cell Transcriptomics	2025
Marklund Lab, Dept. of Medical Biochemistry and Biophysics (MBB), Karolinska Institutet	
<i>Biomedicum C6, Solnavägen 9, 171 65 Stockholm, Sweden</i>	
The aim of the project was to identify regulatory mechanisms in the embryonic ENS by comparing progenitors, neuroblasts and SCPs across regions and timepoints.	
- Finetuned various foundation models (scGPT, UCE, cell2sentence, scPrint2) with custom learning objectives to create region and timepoint classifiers. - Utilized the classifiers to simulate in-silico perturbation of various candidate genes and the attention architecture to study gene-to-gene relations. - Applied SCENIC to infer gene regulatory networks and identify transcription factor motifs. - Integrated scATAC-seq and scRNA-seq data using scGLUE, adding chromatin accessibility evidence to reduce the false positives from sequence-motif inference.	
Biopsy QR Code Management System	2025
Medical Center – University of Freiburg (freelance)	
<i>Hugstetter Straße 55, 79106 Freiburg, Germany</i>	
Developed biopsy management system for Freiburg's medical center. WW	
Benchmarking Single Cell Foundation Models (FM) for Retrieval and Image-to-image Translation Models for Prediction of Cellular Morphology	2023-2025
Yu Li's Laboratory, Dept. of Computer Science and Engineering (CSE), CUHK	
<i>Ho Sin-Hang Engineering Building, Shatin, N.T., Hong Kong</i>	
Benchmarking traditional methods and FMs across diverse biological tasks.	
- Benchmarked retrieval accuracy, diversity, biomarker identification, and novel cell type detection across multiple platforms, species, and technologies. - Evaluated and compared traditional methods with foundation models (scGPT, Geneformer, scMulan, UCE, SCimilarity, scFoundation), designing benchmarks for complex scenarios with ambiguous or incomplete ground truth. - To benchmark MorphDiff established baseline using the DMIT and DRIT++ models. - Adapted the model architectures to process 5 channel cellular morphology imaging data.	
Internship at Centre for Perceptual and Interactive Intelligence (CPII)	2023
InnoHK Centre for Perceptual and Interactive Intelligence	
<i>Building 17W, Hong Kong Science Park, Pak Shek Kok, N.T., Hong Kong</i>	
AI research company targeting sectors of healthcare, urban services, and reindustrialization	
- Developed a prototype for speech-based interaction with LLMs without relying on ASR or TTS. - Utilized speech representations from Hubert pretrained model to finetune LLAMA.	
Internship at ONEFi (iPYGG)	2022-2023
One Financial Technology Limited	
<i>InnoCentre, 72 Tat Chee Avenue, Kowloon Tong, Hong Kong</i>	
Fintech company aiming to connect Asia-Pacific banking into a single application.	
Developed ML models for transaction classification and description summarization.	

PREPRINTS AND JOURNAL PUBLICATIONS

- Resolving cell lineages and gene functions in the developing mouse gastrointestinal tract using in utero transduction** [\[paper\]](#) 2026
Ziwei Liu, Krishnanand Padmanabhan, Jingyan He, Katrin Hector, Bettina Semsch, Jia Sun, Viktoria Knoflach, Sarantis Giatrellis, Johan Lorentz, [Khachatur Dallakyan](#), Christian Göritz, Emma Rachel Andersson, Ulrika Marklund
- SCMBench: Benchmarking Domain-specific and Foundation Models for Single-cell Multi-omics Data Integration** [\[paper\]](#) 2026
Yixuan Wang, Yimin Fan, Xuesong Wang, Tingyang Yu, Yongshuo Zong, Xinyuan Liu, Meitong Liu, Qing Li, Kin hei Lee, [Khachatur Dallakyan](#), Junjie Huang, Gengjie Jia, Jiao Yuan, Ting-Fung Chan, Xin Gao, Irwin King, Yu Li
- Prediction of Cellular Morphology Change Under Perturbations with Transcriptome Guided Diffusion Model** [\[paper\]](#) 2025
Xuesong Wang, Yimin Fan, Yucheng Guo, Chenghao Fu, Kinhei Lee, [Khachatur Dallakyan](#), Yaxuan Li, Qijin Yin, Yu Li, and Le Song

EXTRACURRICULAR ACTIVITIES

- Website for Gohar Petrosyan, ceramicist and painter (Hatsik)** 2026
 - Built and deployed her first online portfolio of clay decorations, ceramic crafts and paintings.
- KI-Mayo 30th annual scientific meeting** 2024
 - Helped to organize main and breakout sessions for “AI and health” section.
- Videogame modding** 2024
 - Modified WW2 videogame resource trading algorithm for AI to prioritize historical allies and trade partners.
- Armenian Red Cross Society (ARCS)** 2019-2020
 - Volunteered as part of the “Social Support and Health Care” team.
- Global Innovation Forum** 2019
 - Volunteered for GIF during Aurora Humanitarian Initiative
 - Helped to organize a talk about the scientific computing framework Torch and its use cases.
- Competitive programming** 2017-2019
 - Codeforces: <https://codeforces.com/profile/khachdallak>
 - Developed and tested competitive programming problems for Armenian SPOJ: <https://am.spoj.com/problems/TIGRANMEC/>

ACHIEVEMENTS & CERTIFICATES

- Introduction to openEHR** 2025
 - Certificate related to understanding of openEHR data modeling, architecture, and application in electronic health records.
- The Swedish Institute Scholarship for Global Professionals** 2024
 - Awarded for demonstrated academic excellence and strong leadership potential.
- CUHK Data Hack 2024** 2024
 - Champion team during the Data Solutions for Social Good Hackathon.
 - Implemented random forest algorithm for early detection and prevention of drug abuse.
- “Belt and Road” scholarship** 2020
 - Scholarship for outstanding new students from the Belt and Road Region countries.
- “Best student” state prize** 2018
 - Republic of Armenia State Prize in Information Technologies.
- “Algorithmization Methods” Programming Contest** 2017
 - Second and third places (top 0.5%) in Autumn and Winter Cup contests respectively.